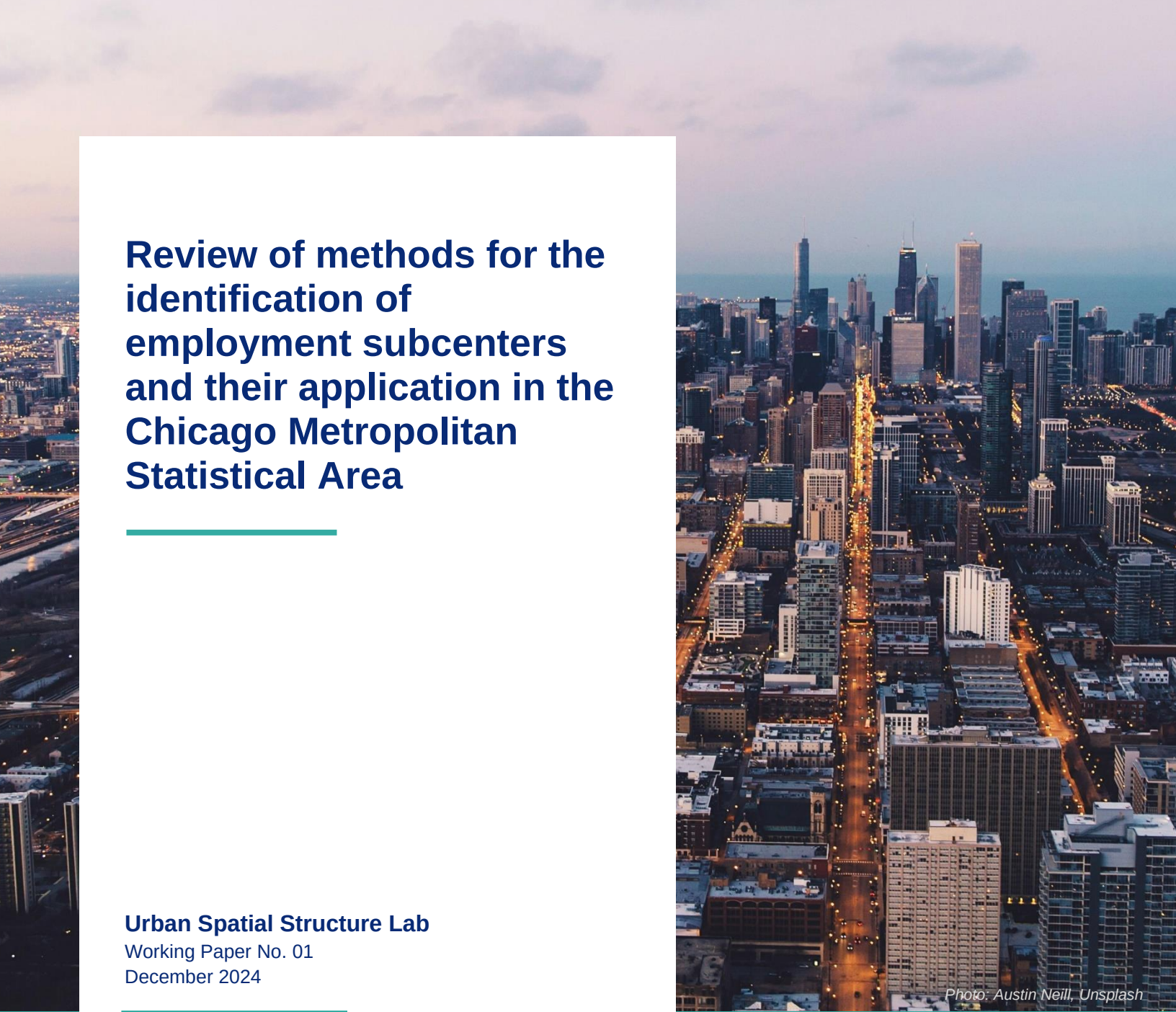




Review of methods for the identification of employment subcenters and their application in the Chicago Metropolitan Statistical Area

Urban Spatial Structure Lab

Working Paper No. 01
December 2024

David López-García
Assistant Professor

Matthew D. Wilson
Associate Director of Economic and Workforce Development, GCI

Kabir Jain
MUPP Candidate '26

Maximiliano Man
MUPP Candidate '25

Nooshin Javidi
PhD Student, MUPP

Elham Pilegar
Research Specialist, GCI

Photo: Austin Neill, Unsplash



Center for Urban
Economic Development



Great Cities Institute

Review of methods for the identification of employment subcenters and their application in the Chicago Metropolitan Statistical Area¹

David López-García, Matthew D. Wilson, Kabir Jain, Maximiliano Man, Nooshin Javidi, Elham Pilegar

Abstract

This article supports the study of urban spatial structures in urban planning by providing a step-by-step guide for applying five distinct methods to identify employment subcenters: double threshold, commuting flows, density peaks, positive residuals, and spatial autocorrelation. We apply these methods to a single study: the Chicago MSA. The results show that applying a distinct method to the same data and area yields highly variable results. We include discussions on selecting the appropriate method and limitations regarding the modifiable area unit problem, the arbitrariness of criteria, the assumption of monocentricity, and identifying the boundaries of employment subcenters.

Introduction

The term "urban spatial structure" refers to the arrangement and interaction of land uses within an urban area (Anas et al., 1998; Kloosterman & Musterd, 2001; Krehl, 2015; Parr, 2014). Studying urban spatial structure involves analyzing the spatial distribution of economic clusters, population density, land uses, built environments, commuting patterns, and the transportation networks that link residents to hubs of economic activity (Cho et al., 2008; Krehl, 2015; Li & Li, 2024; Parr, 2014). Understanding how urban spatial structures evolve is crucial because they impact residents' access to jobs, services, shopping areas, infrastructure, and amenities (Eberts & McMillen, 1999; Farber & Li, 2013). These factors can influence people's residential choices (Cho et al., 2008; Levinson & Kumar, 1994). The urban spatial structure mirrors the locational characteristics of economic activities in a region (Parr, 2014) and has significant implications for urban experiences and overall equity within cities (Knaap et al., 2016; López-García, 2023; López-García & Gómez-Álvarez, 2022). Since the 1980s, empirical research on urban spatial structure has primarily focused on identifying employment

subcenters (Anderson & Bogart, 2001; Cervero & Wu, 1997; Giuliano & Small, 1991; Gordon et al., 1986; McDonald, 1987). While definitions vary according to the method used, an employment subcenter can be broadly defined as an area or a cluster of areas within an urban region that concentrates employment in a more salient way than surrounding areas (Giuliano & Small, 1991; McMillen, 2001).

There is a longstanding tradition linking the identification of employment subcenters with urban planning and policy (Knaap & Rey, 2024; Yu et al., 2021). Concerns about urban sprawl and fragmented growth in metropolitan regions in the United States have sparked significant research into the relationship between housing and labor markets. This research has informed urban planning and policy debates surrounding zoning, affordable housing, environmental regulations, and transit investments (Cho et al., 2008; Knaap & Rey, 2024; Levinson & Kumar, 1994). Urban planners and decision-makers are increasingly called upon to understand how these spatial structures shape cities (Bertaud, 2018; Chen et al., 2019; Krehl, 2015; Parr, 2014). Identifying employment subcenters provides essential building blocks for various urban planning

¹ This research was possible thanks to the support of two rounds of funding from the Gao-Zhang Fund for Geospatial Technologies Research Award allocated to the Urban Spatial Structure Lab at UIC.

and policy studies. For instance, transportation planners can assess the relationship between spatial structure and planning issues such as connectivity between residential and employment areas, establishing public transportation to reduce car dependency, and ways to concentrate development patterns to promote sustainable development (Hu et al., 2020; Farber & Li, 2013; Sharma & Patil, 2024; Xiao et al., 2022). Similarly, economic development experts can investigate whether implemented policies have successfully contributed to the emergence of new employment subcenters, the growth of employment in existing ones, or assessing spatial concentrations of industries or job clusters to compare inter-regional competitiveness and specialization (Agarwal, 2015; Giuliano & Small, 1999). Planners increasingly recognize that spatial structure profoundly impacts individual daily life, social equity, economic growth, and sustainable development.

Despite their significance, the methods for identifying employment subcenters remain rooted mainly in economic and regional development disciplines and are rarely included in urban planning curricula. This omission creates a professional gap, as urban planners often lack the technical knowledge to identify and analyze employment subcenters. Bridging this gap is essential for enabling planners to integrate subcenter identification into broader urban planning initiatives and policy-making processes.

This article aims to support the study of urban spatial structures in urban planning and policy by introducing the most common methods used in the literature to identify employment subcenters. We present five families of methods: double threshold (Giuliano & Small, 1991), commuting flows (Veneri, 2013), density peaks (Craig & Ng, 2001), positive residuals (McDonald & Prather, 1994; McMillen, 2001), and spatial autocorrelation (Baumont et al., 2004; Guillain et al., 2006). Each of these methods has significant variations; however, we will focus on basic versions of these methods to provide a base of knowledge for understanding their more complex variations. While the mathematical components of some methods are pretty simple—primarily involving ratios, averages, and standard deviations—others can be more intricate, relying on regression analysis and measures of spatial autocorrelation. Taking one or two graduate courses in multivariate analysis and

spatial econometrics would be beneficial for mastering the techniques presented in this article.

This step-by-step guide results from two years of work of the Urban Spatial Structure Lab at the University of Illinois Chicago (UIC). The Lab developed a script using R programming language to apply the five distinct methods to the same data and case study: census tract-level employment statistics in the Chicago Metropolitan Statistical Area (MSA). The counties included in the MSA include Cook, DuPage, Grundy, McHenry, Will, DeKalb, Kane, Kendall, and Lake counties in Illinois; Jasper, Lake, Newton, and Porter counties in Indiana; and Kenosha County in Wisconsin. The unit of analysis for each method is the census tract. We use data on total employment for 2021 from the Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) provided by the United States Census Bureau. We link this data with the legal boundaries provided by the U.S. Census Bureau through 2021 TIGER/Line shapefiles. All the spatial layers produced by this study and the R scripts have been made open access through a GitHub repository, which is available upon request. The R script is easily transferable to any other case study, and we provide detailed instructions on adapting it.

The article is structured as follows. The next sections introduce five distinct families of methods. Each family is presented in its sub-section, which includes the context and background in which the method was developed, a step-by-step guide on applying it, and the results we obtain when applying the method to the Chicago MSA. We chose not to include a traditional literature review section; instead, we provide context and background at the beginning of each method sub-section. Following the methods section, we will discuss the issues encountered when applying these five methods to the same case study. This discussion addresses how analyzing the same data with various methods can yield different results and the advantages and disadvantages associated with each method. We include a discussion about selecting the appropriate method for specific urban planning and policy studies. We also highlight challenges in the methods, such as the modifiable area unit problem, the arbitrariness of criteria, the assumption of monocentricity, and

limitations in identifying the boundaries of employment subcenters.

Double Threshold

Sometimes also called the minimum cut-off point method, the double threshold method was initially proposed by Giuliano and Small (1991) in the early nineties to identify employment subcenters in the Los Angeles region. While prior studies had documented the presence of employment subcenters through various methods— from density functions to employment-population ratios, density maps to minimum office floor space—those methods would usually obtain vastly different results even for the same metropolitan area and frequently failed to delimit subcenter boundaries. Determined to bring consistency to the field, Giuliano and Small (1991) proposed the double threshold method to establish a consistent definition that could be applied to identify employment subcenters systematically (Giuliano & Small, 1991).

Giuliano and Small defined an employment subcenter as “a continuous set of zones, each with density above some cutoff point D , that together have at least E total employment and for which all the immediately adjacent zones outside the subcenter have density below D .” (1991, 166). The double threshold method consists of applying two steps based on this definition. The first step is setting up two thresholds, and the second is identifying contiguous geographic units with a common boundary of at least 0.25 miles. For the first step, two thresholds must be specified—one for the absolute number of jobs and another for job density— and the analysis units must be identified that surpass both. The thresholds established in their original study were minimum total employment of 10,000 and a density cutoff of 10 employees per acre (Giuliano & Small, 1991). One aspect of the method subject to considerable debate is how and where to locate the thresholds because the method has proven to be highly sensitive to the choice of cut-off value (Yu et al., 2021). A group of Latin American scholars explored and successfully applied an approach where the thresholds are established based on the statistical distribution of employment in the urban region under analysis (Aguilar & Hernández, 2012;

Garrocho & Campos, 2007; López-García & Gómez-Álvarez, 2022). In this approach, the thresholds are set through the following criteria:

- 1) An employment magnitude above the mean of the urban area under study plus one standard deviation.
- 2) An employment density above the mean of the urban area under study.

Following Garrocho and Campos (2007), we use the following formula to identify the units of analysis that exceed the two thresholds:

$$\begin{aligned} 1) \quad M_i &> (\bar{M}_c) + (SD \bar{M}_c) \\ 2) \quad D_i &> \bar{D}_c \end{aligned}$$

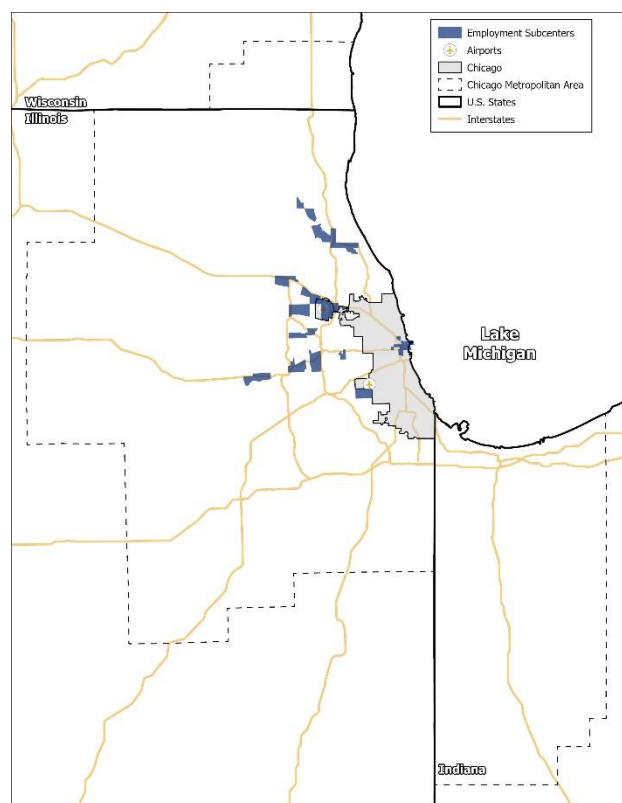
Where M is employment magnitude (absolute number of jobs), i is the unit of analysis, c is the urban area under study, D is employment density (absolute number of jobs/Area of the unit of analysis), SD is standard deviation, $\bar{M}$ is average employment magnitude, and $\bar{D}$ is average employment density. All the units of analysis that surpass the two thresholds concentrate employment, but not all of them will be considered part of an employment subcenter. Identifying the employment subcenters needs one more step.

The second step of the method is to identify the clusters of units of analysis that, having surpassed both thresholds, are contiguous and share a common boundary of at least 0.25 miles (Giuliano & Small, 1991). Only the contiguous clusters of observation units will be classified as employment subcenters. The units of observation that are isolated will be discarded from the analysis. The boundaries of the clusters of units of analysis should then be merged, and each of them will be considered an employment subcenter on its own. The jobs within each of the original units of analysis can be added and attributed to the employment subcenter. Once identified, the subcenters can be classified in a hierarchy according to their absolute number of jobs.

Applying the method to the Chicago MSA yields the employment magnitude threshold at 9,249 jobs and the density threshold at 12.2 jobs per hectare. The resulting employment subcenters are shown in Map 1, distributed in the Loop and along the main

highways in the MSA. The double threshold method allows us to identify eleven employment subcenters in 2021. With a combined 1,171,669 total jobs, the eleven employment subcenters concentrate 26.9 of the jobs available in the Chicago MSA in 0.88 percent of its territory. The main employment subcenter is the Loop. With 690,582 jobs in an area of 1,433.5 Hectares, the Loop concentrates 15.9 of the jobs in the metropolitan region in only 0.06 percent of its territory. The Loop is the densest employment subcenter in Chicagoland, with 619 jobs per Hectare.

Map 1. Employment subcenters identified through the double threshold method, 2021.



Density Peaks

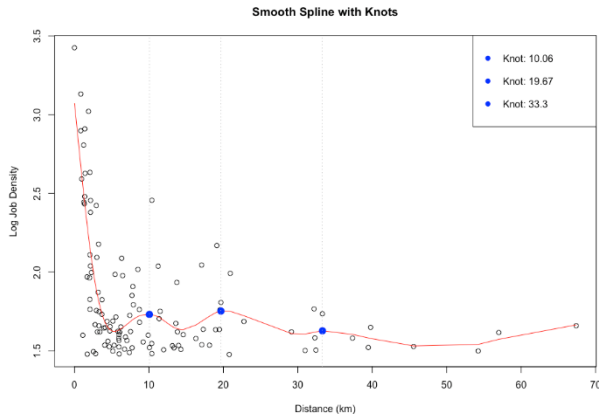
The first iteration of the density peaks method was proposed by McDonald (1987) and McDonald and McMillen (1990). The authors examined employment density in Chicago in 1956 and 1970 and argued that “a zone shall be considered an employment subcenter if its value for the measure in question exceeds the value for each of the contiguous zones” (McDonald & McMillen 1990, 1562). A few years

later, Craig and Ng (2001) highlighted a shortfall of McDonald and McMillen’s approach: identifying local employment peaks had not considered other relevant covariates of employment formation, such as distance from the primary employment concentration in an urban region.

Building on McDonald and McMillen’s (1990) pioneering efforts, Craig and Ng (2001) proposed a revised version of the density peaks method based on nonparametric estimation of a quintile smoothing spline for the upper tail of the employment density distribution. A smoothing spline is a statistical method that creates a smooth curve to capture the relationship between two variables – in this case, employment density and distance to the CBD. The smoothing spline will be able to show density peaks graphically, which are then considered as the location of the employment subcenters. According to Craig and Ng (2001), their method reduces the need to set up arbitrary cut-off points, like in the case of the double-threshold technique, allowing the development of a statistically based definition of an employment subcenter (Craig & Ng, 2001).

The method entails eight steps. The first is to identify and set apart the upper 95th percentile of the job density distribution of the urban region under analysis. Only the upper 95th percentile of the distribution will be analyzed. The second step is to identify the CBD of the urban region under analysis and calculate the distance to the CBD from the centroid of each unit of analysis. The CBD can be identified through prior knowledge of its exact location. The third step is to produce a scatterplot of the relationship between the distance to the CBD and the logarithm of job density. The fourth step is calculating a smoothing spline, which balances how closely the curve fits the data. Here, Craig and Ng (2001) build on the work of Koenker et al. (1994), who proposed a smoothing parameter called lambda (λ) to find the degrees of freedom in the regression that would produce a curve with the proper equilibrium between smoothness and accuracy. In our case study, the lambda (λ) parameter found the smoother regression curve at 12.1 degrees of freedom. The result is a regression curve with waves through the data indicating local increases in log job density. The peak of each of those waves is called a knot (see Figure 1).

Figure 1. Smoothing spline with knots.

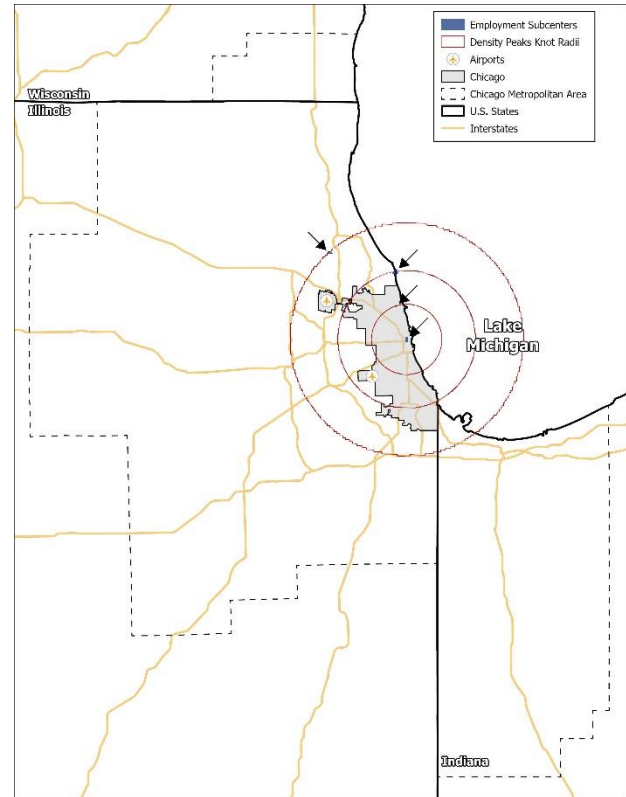


The fifth step is identifying the knots in the smoothing spline. Each knot indicates the place in the regression curve where job density peaks before it starts decreasing again. The sixth step is to identify the unit of analysis linked to each knot and its exact distance from the CBD. This can be done by drawing vertically dotted lines in each knot and connecting the units of analysis with the x-axis of the scatterplot. The place where the dotted lines intersect with the x-axis indicates the distance from the CBD. Figure 1 shows a scatterplot of the logarithm of job density and distance to the CBD. The figure also shows the smoothing spline, the knots, and a straight line from each knot to the x-axis, indicating the distance from the CBD where employment subcenters are located. The seventh step must be done on a map. For each knot, draw a buffer of its distance from the CBD. These buffers are called subcenter rings. The eighth and final step is to look at each subcenter ring and identify the unit of analysis with the highest job density that intersects the ring. It is this intersection that indicates the exact location of an employment subcenter.

Map 2 shows the subcenter rings for each of the knots. In addition to the CBD, one of the employment subcenters is located in northern Chicago (subcenter ring 1), another one in Evanston (subcenter ring 2), and the last one north of O'Hare airport (subcenter ring 3). One of the main limitations of this method is that it tells us the location of an employment subcenter. However, it fails to show us the total number of units of analysis that might be part of each employment subcenter and its boundaries. In any

case, the four census tracts identified by this method concentrate 332,089 jobs, which translates into 7.6 percent of existing jobs in the MSA.

Map 2. Employment subcenters identified through density peaks method, 2021.



Positive Residuals

The statistical approach to identifying departures from a monocentric employment density model was first introduced by McDonald and Prather (1994) and applied to identify suburban employment centers in the Chicagoland region. This method utilizes a monocentric regression model of employment density as a function of distance to a CBD. Employment subcenters are detected by identifying significantly positive residuals from the regression model. The method was developed in response to the perceived arbitrariness of researcher-defined cutoffs and thresholds utilized to detect employment subcenters and prior research that treated subcenters as distinct units rather than peaks in a continuous employment density function of a region (McDonald and Prather 1994). Since the positive residual method of employment subcenter detection

is based on unique density gradients that vary across regions, the method detects subcenters in a manner that is sensitive to local employment landscapes (i.e., different diminishing of employment density as distance from CBD increases) and varying geographic terrains (McDonald and Prather 1994). An additional benefit of the positive residual employment subcenter detection method is that estimation can be applied to many regions where a researcher does not have detailed knowledge or a priori expectations of the employment landscape.

This method starts with fitting a simple regression model of employment density as a function of distance to a CBD for each geographic unit of a region. This monocentric employment density model can be estimated as

$$\log(y_i) = \beta_0 + \beta_1 \text{DistCBD}_i + \varepsilon_i$$

where $\log(y_i)$ is the natural log of employment density for geographic unit i , DistCBD_i is distance to the CBD for geographic unit i , and ε_i is an error term. Employment subcenters can be identified through examining the residuals from the regression model and identifying significantly positive residuals where geographic units that are greater than or equal to an upper confidence limit. This can be estimated as

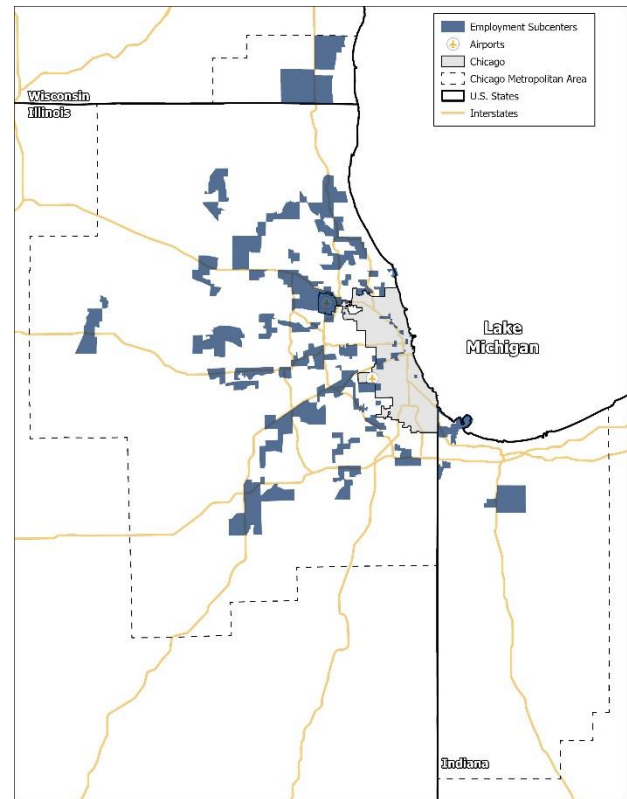
$$(y_i - \hat{y}_i) / \hat{\sigma}_i > c$$

where $\hat{y}_i$ is the predicted value of employment (y) at geographic unit i ; y_i is the observed value ($y_i - \hat{y}_i$) is the residual value for geographic unit i ; $\hat{\sigma}_i$ is the estimated standard error of prediction i ; and c is the critical value for a normal distribution (in our case, critical values associated with 5 percent, 10 percent, and 20 percent significance levels are 1.96, 1.64, and 1.28). This estimation strategy described above most closely follows McDonald and Prather (1994).

Adding onto this method, McMillen (2001, 2003) uses the positive residual method from McDonald and Prather (1994) as a first stage to identify subcenter candidates and includes a second stage to assess the significance of the candidate subcenters. The second stage uses semiparametric regression to assess the influence of candidate subcenters on the regional employment density

function. This involves utilizing a flexible, nonparametric approach to model the impact of distance to the central business district (CBD) while testing the effects of distances to candidate subcenters. Additionally, McMillen (2001, 2003) incorporates geographic weights to address spatial autocorrelation issues when employment densities in neighboring geographic units might be spatially autocorrelated. This refinement addresses spatial dependencies that units have with their neighbors through the spatial clustering of employment. This process distinguishes between subcenters with localized impacts and ones that have a larger regional influence. A stepwise estimation procedure refines the model by selecting only statistically significant variables (see McMillen, 2001). Lastly, to identify the boundaries of employment subcenters, and in line with the criteria used in McMillen (2003), we consider an employment subcenter the tracts with statistically significant positive residuals and that share a common boundary of at least 0.25 miles (Giuliano & Small, 1991).

Map 3. Employment sub-center identification through Positive Residuals Method, 2021.



Map 3 shows the resulting employment subcenters when applying the positive residuals method with geographic weights to the Chicago MSA. The spatial weights are defined using an optimal bandwidth function called `gwr.sel` in the R-package `spData` (more information about spatial weights is in the next section). The method identifies 40 employment subcenters with almost 1.7 million jobs. The employment subcenters identified by this method concentrate 41.7 percent of existing jobs in the MSA in 6.8 percent of its territory. In line with other monocentric approaches applied to the metro area, the identified CBD is roughly in downtown Chicago. Employment subcenters highlighted on the map consist of clusters of more than two contiguous Census Tracts that have job numbers that deviate significantly positively from the expected number based on the modeled relationship between distance from the CBD and the number of jobs.

Spatial autocorrelation

Using spatial autocorrelation to identify employment subcenters was first proposed by Baumont et al. (2004) and refined by Guillain et al. (2006). The motivation behind Baumont et al. was to develop an alternative identification methodology that constituted “an improvement over existing methodologies that do not assess the significance of their results and necessitate the definition of arbitrary cut-off points” (2004, 147). Instead of having to choose arbitrary cut-off points, spatial autocorrelation techniques take advantage of the spatial data in the urban region under study which are characterized by two spatial effects: spatial autocorrelation and spatial heterogeneity (Guillain et al., 2006). In this way, this method is based on analyzing the inherent structure of spatial data to identify spatial clusters with significantly similar and prominent socioeconomic, physical, or activity features based on relevant indicators of economic activity (Yu et al., 2021).

The method can be summarized in five steps. The first step is to determine a spatial weight matrix for the analysis to model the spatial interdependence between the observation units. Here, the method needs researchers to somewhat arbitrarily decide how the observation units are connected to a set of

neighboring units of observations. Various spatial weight matrices have been used in the literature: simple binary contiguity matrices, binary spatial weight matrices with a distance-based critical cutoff above which spatial interactions are assumed negligible, and generalized distance-based spatial weight matrices (Chi and Zhi 2020, 35). Choosing a specific weight matrix is still one of this method's most challenging and controversial methodological issues. Both Baumont et al. (2004) and Guillain et al. (2006) used nearest-neighbor matrices.

The second step is to choose the appropriate indicator to be analyzed. Again, researchers need to decide arbitrarily which indicator to use. Baumont et al. (2004) analyzed two indicators separately: total employment and employment density. Guillain et al. (2006) analyzed the employment-to-population ratio. The third step is to test spatial autocorrelation in the area under study. Here, Moran's I is used to test for global spatial autocorrelation between all units of analysis. If significant, this means that areas with similar values (high or low) of the indicator under analysis tend to be spatially clustered.

The results from the previous step need to be refined to identify the spatial clustering of high values in the variable under study. This gives way to the fourth step, testing local autocorrelation within the sample. Different spatial autocorrelation statistics exist. For instance, there is the Local Index of Spatial Autocorrelation (LISA) and Hot Spot Analysis (Getis-Ord G_i^*). Both Baumont et al. (2004) and Guillain et al. (2006) use the Local Indicator of Spatial Associations (LISA), and so do we. Testing for local spatial autocorrelation needs a couple of procedures. Based on a Moran scatterplot plotting the spatial lag w_z against the original values z of a variable, the results of the LISA will allow the classification of each unit of analysis in one of four types of spatial association with its neighbors: High-High (HH), Low-High (LH), High-Low (HL), and Low-Low (LL). The method requires an assessment of the statistical significance of such spatial associations. Therefore, only the significant HH and HL units of analysis should be considered employment subcenters (Guillain et al., 2006). Local Indicators of Spatial Association (LISA) are used for this purpose through the following equation:

$$I_i = \frac{(x_i - \mu)}{m_0} \sum_j \omega_{ij} (x_{ij} - \mu)$$

$$\text{with } m_0 = \frac{\sum_i (x_i - \mu)^2}{n}$$

Where I_i is the local version of Moran's I statistic for unit i , x_i is the observation in unit i , μ is the mean of the observations across spatial units and where the summation over j is such that only neighboring values of j are included. A positive value for I_i indicates spatial clustering of similar values (HH or LL). In contrast, a negative value indicates spatial clustering of dissimilar values (HL or LH) between a zone and its neighbors (Baumont et al., 2004; Guillain et al., 2006). The fifth and last step is to delimit the employment subcenters. Only the significant HH and HL clusters of observation units should be considered employment subcenters. A set of significant HH units of observation indicates an economic center covering several neighboring units of observation, while significant HL units of observation represent isolated centers.

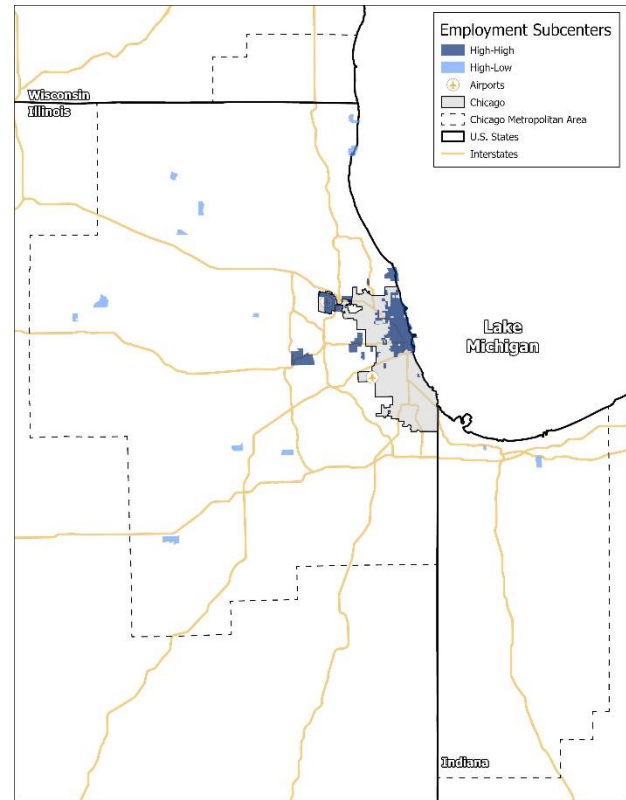
We analyzed the case of Chicago using a k-nearest neighbors approach for the weight matrix, with seven nearest neighbors chosen based on the average number of neighbors for each tract in the study area. We started by testing global spatial autocorrelation. The Moran's I test result shows a statistically significant spatial clustering of employment density for 2021 (Table 1). The positive Moran's I value (0.486), along with the high standardized score (46.22) and significance at the 5% level, indicates that areas with similar employment density levels (either high or low) are spatially grouped. This finding implies that employment density is unevenly distributed and exhibits a spatial pattern.

Table 1. Moran's I Statistics for Employment Density, 2021.

7 nearest neighbors weight matrix			
Variable	Moran's I	St. dev.	St. value
Employment Density	0.486***	0.010	46.22

The expected value for Moran's I statistic is -0.0004 for Employment Density. All statistics are significant at the 5% level.

Map 4. Employment sub-center identification through Spatial Autocorrelation method, 2021.



Map 4 illustrates the results of the Local Moran's I analysis for employment density in 2021. The clusters shown represent tracts identified with significant p-values, highlighting areas where employment densities are spatially correlated. The "High-High" areas, marked in blue, indicate regions with high employment densities surrounded by similarly high values, suggesting spatial clustering of economic activity. This means that the spatial autocorrelation method identified 26 employment subcenters with 1.13 million jobs. That is, said employment subcenters concentrate 26 percent of existing jobs in the MSA in 1.1 percent of its territory.

Commuting Flows

This method utilizes commuting patterns to assess the balance between job and residential densities. Ultimately, the method detects areas with high inflows of commuters as an indication that areas have concentrations of employment. Veneri (2013) deployed a commuting flow method to identify polycentric employment subcenters in Rome and

Milan that illustrated how commuting flows provide a more refined approach to distinguishing between areas with local versus regional influence based on commuting patterns. Veneri's (2013) commuting flows employment subcenter detection method relies on three measures to determine if an area can be classified as an employment subcenter. The first is flow centrality (FC). For each unit of analysis, this metric is defined as the ratio of total inbound commuters to total outbound commuters. A flow centrality ratio above 1 indicates a tendency to be a job center, while a ratio below 1 indicates more of a residential community. In other words,

$$FC_i = \frac{Inflow_i}{Outflow_i}$$

The second is Directional Dominance Index (DDI). The directional dominance of a unit of analysis is defined as the number of inbound commuters in the unit divided by the average inbound commuters for all units in the metro area. This metric pits units against each other, and values are expected to follow a normal distribution centered on an average of 1.

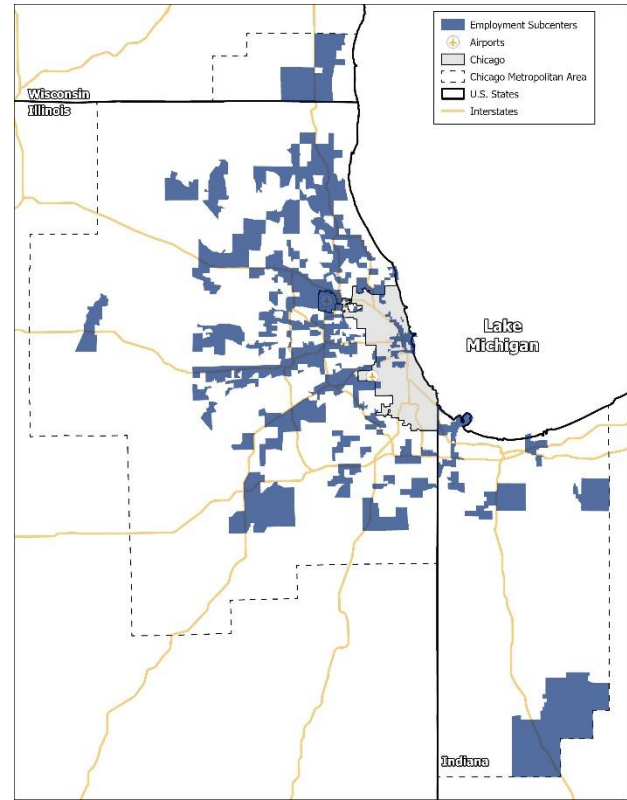
$$DDI_i = \frac{Inflow_i}{\frac{1}{n}(\sum_{k=1}^n Inflow_k)}$$

The third is Productive Completeness (PC). This metric measures the diversity of the existing jobs for each unit of analysis. While productive completeness is not an indicator of the concentration of employment per se, it does provide a measure of the concentration of economic sectors in a unit of analysis. In our analysis, we use the North American Industry Classification System (NAICS), but other classification systems might exist in other case studies. In our case, productive completeness for a unit of analysis can be defined as the number of distinct 2-digit NAICS sectors with non-zero employment within the unit divided by the average number of distinct 2-digit NAICS sectors with non-zero employment in every unit of analysis across the metro region.

$$PC_i = \frac{(DistinctNAICS_i)}{\frac{1}{n}(\sum_{k=1}^n DistinctNAICS_k)}$$

In each of the equations above, i refers to the current unit of analysis. Once these three measures are calculated for each unit of analysis, the census tracts with a value above 1 in all three indicators are selected as subcenters. To ensure comparability with the rest of the methods, we applied contiguity criteria and considered employment subcenters only the tracts with a common boundary of at least 0.25 miles (Giuliano & Small, 1991).

Map 5. Employment subcenters identified by the commuting flows method, 2021.



Map 5 shows the results when applying the commuting flows methodology to the Chicago MSA. The method identified 35 employment subcenters distributed in Chicago's downtown and adjacent areas along the network of interstate highways. With a total of 2.75 million jobs, the employment subcenters identified through this method concentrate 63.1 percent of the existing jobs in the MSA in 13.8 percent of its territory.

Discussion

Table 2 summarizes the results from applying the five methods to the case of the Chicago MSA. As the table shows, each of the distinct methods produces divergent results. For example, the double threshold method identifies 11 employment subcenters, density peaks 4, positive residuals 40, spatial autocorrelation 26, and commuting flows 35. There are also notable differences in the number of census tracts included in the employment subcenters identified by each method, the total jobs in the employment subcenters, the share of jobs and area in the employment subcenters as a share of the metropolitan statistical area, and job density in the employment subcenters.

There are also important differences in the spatial distribution of the employment subcenters and the ability to delimit their boundaries. Delimiting boundaries allows calculating key characteristics of employment subcenters like the total amount of jobs

within them, their rank in the urban hierarchy, their economic specialization, and assessing if the area under analysis trends towards a polycentric or a monocentric urban form. Delimiting the boundaries is also a helpful layer of information for research projects where the exact location of employment subcenters constitutes an important input. In this study, we used the contiguity criteria to delimit the boundaries of employment subcenters in four methods: double threshold, commuting flows, positive residuals, and spatial autocorrelation. The maps produced by these three methods show that employment subcenters are concentrated in downtown areas, around O'Hare airport, and along interstate highways. The case of the density peaks method is different. Through the concentric rings, we can point to the location of an employment subcenter. However, the method does not allow us to identify all of the units of analysis that might be part of said employment subcenter.

Table 2. Summary of employment subcenters identified through five distinct methods.

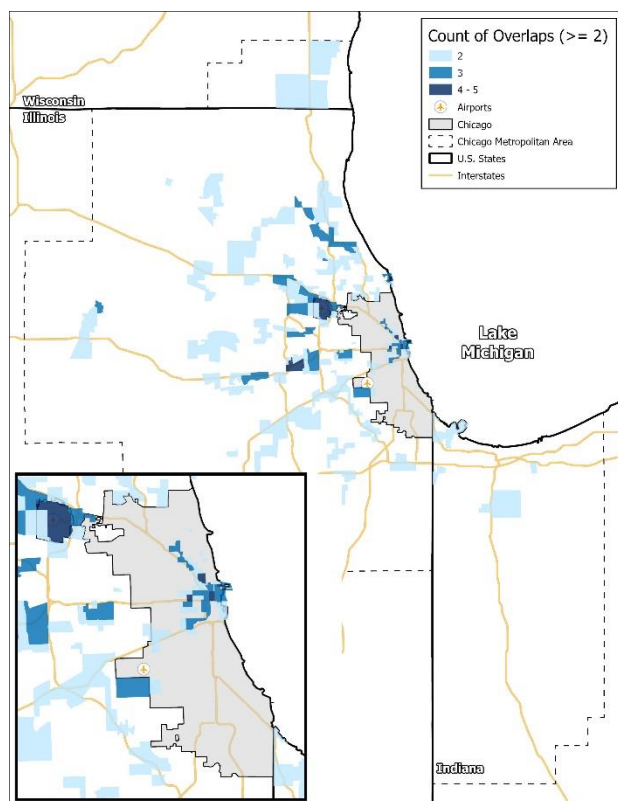
Method	No. of Employment Subcenters (ES)	No. of Census tracts in ES	Total jobs in ES	Area of ES (Ha)	Jobs as a share of MSA (%)	Area as a share of MSA (%)	Job density in ES (jobs/Ha)
Double Thresholds	11	45	1,171,669	21,878.0	26.9	0.9	53.6
Density Peaks	4	n/a	n/a	n/a	n/a	n/a	n/a
Positive Residuals	40	175	1,815,205	168,949.5	41.7	6.8	10.7
Spatial Autocorrelation	26	288	1,132,636	26,564.8	26.0	1.1	42.6
Commuting Flows	35	383	2,749,623	341,040.8	63.1	13.8	8.1

Map 6 shows a count of overlaps in the units of analysis in terms of how many methods identified them as part of an employment subcenter. As the map shows, only three locations are identified as employment subcenters by four out of four methods: Downtown Chicago and adjacent areas, O'Hare Airport, and Evanston. Three methods identified a few areas, mainly in Downtown Chicago and its adjacent areas along one specific Interstate Highway. Many units of analysis were identified as employment subcenters by two methods along the array of interstate highways in the MSA.

An aspect that deserves careful consideration is deciding which method is the right one to use for specific studies. Here, the literature suggests that the selection of the method depends on the research goals, data availability, the spatial scale of the analysis, and the local context of the city under study (Aguilar & Hernández, 2012; López-García & Gómez-Álvarez, 2022). When the purpose of the study is to compare the evolution of urban spatial structure in one city across several points in time, using methods based on cutoff points, such as the double threshold method, is recommended. When the goal is to compare urban spatial structure among

a sample of cities, methods based on spatial econometric estimations such as positive residuals, density peaks, or spatial autocorrelation are recommended. A comparison of cities would benefit from an econometric estimation method because each city's subcenters will be identified based on the city's employment dynamics. A valuable lesson from this research project is that a *priori* knowledge of the case study matters. The methods that identify employment subcenters as a function of distance from the CBD work better in a case study with a monocentric urban structure. That is, the positive residuals and the density peaks method might work better in a case study like Chicago, which is highly monocentric. In contrast, the methods that do not rely on the assumption of monocentricity, such as double threshold, spatial autocorrelation, and commuting flows, might work better for polycentric cities where the influence of the CBD is decreasing.

Map 6. Overlap of methods in Chicago MSA, 2021.



One of the most debated aspects of identifying employment subcenters is the number of discretionary decisions researchers must make. Each iteration of methodological innovation seems to be motivated by

reducing the number of discretionary decisions that must be made by the researchers and pursuing objectivity. However, there remain inescapable elements in all the methods that need a degree of discretionary decision. For example, based on the establishment of cut-off points, the double threshold method was highly criticized for its subjectivity in deciding about the cut-off points. In their original article, Giuliano and Small (1991) located the cut-off points at 10,000 jobs and ten employees per acre. In a subsequent paper, they moved the cut-off points to 20,000 jobs and 20 employees per acre (Giuliano et al., 2007). Aiming to overcome this subjectivity in the double threshold method, scholars have developed approaches to establishing the cut-off points based on the statistical distribution of employment in the area under study (Garrocho and Campos 2007; Aguilar and Hernández 2012). Others have proposed adaptations of the double threshold method where the cut-off points decline with distance from the CBD (Ban et al., 2017). Even the methods developed to deal with the discretionality of the double threshold method have some aspects that must be decided subjectively. For example, the positive residuals method requires the researchers to make a discretionary decision about the neighboring matrices used in the geographically weighted regression (McMillen, 2001). Furthermore, when researchers apply the positive residuals method and wish to delimit the boundaries of employment subcenters, they must resort to the establishment of discretionary cut-off points (McMillen, 2003). The spatial autocorrelation method does not escape discretionality either. One of the fundamental steps of the method is establishing a weight matrix to model the spatial interdependence between the observation units (Baumont et al., 2004; Guillain et al., 2006). There are different ways to establish a weight matrix in spatial econometrics, and selecting the appropriate criteria continues to be one of the more debated aspects of the spatial autocorrelation method.

Another aspect of the methods that deserves considerable attention is the modifiable areal unit problem, which refers to the arbitrary selection of the units of analysis. In spatial analysis and geography, the modifiable areal unit problem argues that statistical analysis results can be affected by how geographic units (areas) are defined or aggregated (Chi & Zhu, 2020). The literature on employment

subcenter identification shows that the use of different units of analysis significantly influences the results of the methods (Knaap & Rey, 2024; Yu et al., 2021). In our case, we ran the methods using distinct geographic units of analysis, including grids of 1 kilometer, 1 mile, and 1 hectare, as well as administrative boundaries like the census block, census group, census tract, and zip code. We confirmed that applying the same method to the same employment data using distinct geographical units of analysis produces dismal results. In general, we found that a smaller unit of analysis produces a more scattered spatial pattern with more employment subcenters. Conversely, a bigger unit of analysis produces a more concentrated spatial pattern with less employment subcenters. For this article, we decided to present the results using the census tract because it is the most common unit of analysis used in literature and because of the legibility of the maps produced at this geographical unit. Nevertheless, it was an arbitrary decision.

Finally, it is noteworthy to reflect on the historical and theoretical roots of the concepts of employment “center” and subcenter” and how those concepts shape our understanding of urban spatial structure. The concept of a “subcenter” inherently suggests a hierarchical urban structure with a primary “center” at the top. This concept aligns with the assumptions of monocentric urban models, which posit that cities are structured around a single dominant core that serves as a focal point for a region, and subcenters are secondary to the primary center. By contrast, a polycentric region could have multiple centers with no dominant single core, and calling these independent centers subcenters may obscure their formation and existence.

Additionally, “employment centers” and “subcenters” suggest a particular way of understanding and organizing urban space. These constructs are grounded in economic geography and regional science logic, which prioritize measurable metrics like density, commuting flows, or employment concentrations to define centers. However, this emphasis on quantitative indicators may overlook other forms of centrality, such as cultural, social, or historical significance, which are harder to quantify but equally vital in shaping urban experiences (Jacobs, 1961; Parr, 2004).

The debate between the Chicago School and the Los Angeles School of urban theory provides a useful lens for understanding the implications of these constructs (Dear, 2003). The Chicago School, rooted in monocentric models, emphasizes a hierarchical urban structure where subcenters are extensions of a dominant CBD (Alonso, 1964). This perspective is reflected in planning practices that prioritize extending the reach of the CBD or managing growth at its edges. In contrast, the Los Angeles School challenges this monocentric view, arguing that cities like Los Angeles exemplify polycentric urbanism. In this framework, multiple nodes of activity operate as independent hubs shaped by economic, social, and cultural forces at various scales (Scott & Storper, 2007; Soja, 2000). Planners must remain critical of how the labels of “center” and “subcenter” are applied to their understanding of regions, where the concepts come from and their potential to oversimplify the complexities of urban landscapes.

Conclusions

This article presented five distinct methodologies for identifying employment subcenters and applying them to the Chicago Metropolitan Statistical Area. The study's findings indicate that using different methods on the same data and case study can lead to significantly varying results. The number of identified employment subcenters ranged from 4 to 74, depending on the methodology employed. Additionally, we assessed the spatial distribution of these subcenters, finding that most are located in Downtown Chicago and its neighboring areas, around O'Hare Airport, Evanston, and in corridors along the interstate highway system. To our knowledge, this is the first study that compares five distinct methods for identifying employment subcenters in the same case study and presents the results in the same article.

We hope our step-by-step guide on applying these methods for identifying employment subcenters will benefit planning education, research, and practice. Understanding urban spatial structure through identifying employment subcenters is a crucial component that planners can utilize in their work. These subcenters can be the focus of planning and policy interventions while also providing valuable contextual information for broader planning efforts. A

clear understanding of urban spatial structure can facilitate discussions related to various fields, such as economic development, workforce development, housing, transportation, land use, infrastructure investment, community development, and spatial

planning. Ultimately, we aim for the application of methods for assessing urban spatial structure to contribute to broader conversations about the urban experience and equity in urban regions.

References

- Agarwal, A. (2015). An examination of the determinants of employment center growth: Do local policies play a role? *Journal of Urban Affairs*, 37(2), 192–206.
- Aguilar, A., & Hernández, J. (2012). Transformación metropolitana y estructura policéntrica en la Ciudad de México. Identificación de subcentros urbanos, 1990-2005. In A. Ziccardi (Ed.), *Ciudades del 2010: Entre la sociedad del conocimiento y la desigualdad social* (pp. 199–237). Universidad Nacional Autónoma de México.
- Alonso, W. (1964). The Historic and the Structural Theories of Urban Form: Their Implications for Urban Renewal. *Source: Land Economics*, 40(2), 227–231.
- Anas, A., Arnott, R., & Small, K. A. (1998). Urban Spatial Structure. *Source: Journal of Economic Literature*, 36(3), 1426–1464.
- Anderson, N. B., & Bogart, W. T. (2001). The structure of sprawl: Identifying and characterizing employment centers in polycentric metropolitan areas. *American Journal of Economics and Sociology*, 60(1), 147–169.
- Ban, J., Arnott, R., & Macdonald, J. L. (2017). Identifying employment subcenters: The method of exponentially declining cutoffs. *Land*, 6(17), 1–33.
- Baumont, C., Ertur, C., & Le Gallo, J. (2004). Spatial analysis of employment and population density: The case of the agglomeration of Dijon 1999. *Geographical Analysis*, 36(2), 146–176.
- Bertaud, A. (2018). *Order without design: How markets shape cities*. The MIT Press.
- Cervero, R., & Wu, K.-L. (1997). Polycentrism, commuting, and residential location in the San Francisco Bay area. *Environment and Planning A*, 29, 865–886.
- Chen, T., Hui, E. C. M., Wu, J., Lang, W., & Li, X. (2019). Identifying urban spatial structure and urban vibrancy in highly dense cities using georeferenced social media data. *Habitat International*, 89(102005).
- Chi, G., & Zhu, J. (2020). *Spatial regression models for the social sciences*. Sage.
- Cho, E. J., Rodriguez, D. A., & Song, Y. (2008). The role of employment subcenters in residential location decisions. *Journal of Transport and Land Use*, 1(2), 121–151.
- Craig, S. G., & Ng, P. T. (2001). Using quantile smoothing splines to identify employment subcenters in a multicentric urban area. *Journal of Urban Economics*, 49(1), 100–120.
- Dear, M. (2003). The los angeles school of urbanism: An intellectual history. *Urban Geography*, 24(6), 493–509.
- Eberts, R. W., & McMillen, D. P. (1999). Agglomeration economies and urban public infrastructure. In E. Mills & P. Cheshire (Eds.), *Handbook of Regional and Urban Economics*. Elsevier.
- Farber, S., & Li, X. (2013). Urban sprawl and social interaction potential: An empirical analysis of large metropolitan regions in the United States. *Journal of Transport Geography*, 31, 267–277.
- Garrocho, C., & Campos, J. (2007). Dinámica de la estructura policéntrica del empleo terciario en el área metropolitana de Toluca, 1994-2004. *Papeles de Población*, 13(52), 110–135.

-
- Giuliano, G., Redfearn, C., Agarwal, A., Li, C., & Zhuang, D. (2007). Employment concentrations in Los Angeles, 1980-2000. *Environment and Planning A*, 39(12), 2935–2957.
- Giuliano, G., & Small, K. A. (1991). Subcenters in the Los Angeles region. *Regional Science and Urban Economics*, 21, 163–182.
- Giuliano, G., & Small, K. A. (1999). The determinants of growth of employment subcenters. *Journal of Transport Geography*, 7, 189–201.
- Gordon, P., Richardson, H. W., & Wong, H. L. (1986). The distribution of population and employment in a polycentric city: the case of Los Angeles f. *Environment and Planning A*, 18, 161–173.
- Guillain, R., Le Gallo, J., & Boiteux-Orain, C. (2006). Changes in Spatial and Sectoral Patterns of Employment in Ile-de-France, 1978-97. *Urban Studies*, 43(11), 2075–2098.
- Hu, L., Yang, J., Yang, T., Tu, Y., & Zhu, J. (2020). Urban Spatial Structure and Travel in China. *Journal of Planning Literature*, 35(1), 6–24.
- Jacobs, J. (1961). *The death and life of great American cities*. Random House.
- Kloosterman, R. C., & Musterd, S. (2001). The polycentric urban region: Towards a research agenda. *Urban Studies*, 38(4), 623–633.
- Knaap, E., Ding, C., Niu, Y., & Mishra, S. (2016). Polycentrism as a sustainable development strategy: empirical analysis from the state of Maryland. *Journal of Urbanism: International Research on Placemaking and Urban Sustainability*, 9(1), 73–92.
- Knaap, E., & Rey, S. (2024). Measuring two decades of urban spatial structure: The evolution of agglomeration economies in American metros. *Computers, Environment and Urban Systems*, 110(102116), 1–18.
- Koenker, R., Ng, P., & Portnoy, S. (1994). Quantile Smoothing Splines. *Biometrika*, 81(4), 673–680.
- Krehl, A. (2015). Urban spatial structure: An interaction between employment and built-up volumes. *Regional Studies, Regional Science*, 2(1), 290–308.
- Levinson, D., & Kumar, A. (1994). The Rational Locator: Why Travel Times Have Remained Stable. *Journal of the American Planning Association*, 60(3), 319–332.
- Li, J., & Li, C. (2024). Characterizing urban spatial structure through built form typologies: A new framework using clustering ensembles. *Land Use Policy*, 141(107166), 1–11.
- López-García, D. (2023). *Worker mobility and urban policy in Latin America: Policy interactions and urban outcomes in Mexico City*. New York and Abingdon: Routledge.
- López-García, D., & Gómez-Álvarez, D. (2022). Urban structure of Guadalajara Metropolitan Area, 1999-2019: An analysis of employment subcenters. *Estudios Demograficos y Urbanos*, 37(2), 427–481.
- McDonald, J. F. (1987). The Identification of Urban Employment Subcenters'. *Journal of Urban Economics*, 21, 242–258.
- McDonald, J. F., & McMillen, D. P. (1990). Employment subcenters and land values in a polycentric urban area: The case of Chicago. *Environment and Planning A*, 22, 1561–1574.
- McDonald, J. F., & Prather, P. J. (1994). Suburban Employment Centres: The Case of Chicago. *Urban Studies*, 31(2), 201–218.
- McMillen, D. P. (2001). Nonparametric employment subscenter identification. *Journal of Urban Economics*, 50(3), 448–473.
- McMillen, D. P. (2003). Identifying sub-centres using contiguity matrices. *Urban Studies*, 40(1), 57–69.
- Parr, J. B. (2004). The polycentric urban region: A closer inspection. *Regional Studies*, 38(3), 231–240.

-
- Parr, J. B. (2014). The Regional Economy, Spatial Structure and Regional Urban Systems. *Regional Studies*, 48(12), 1926–1938.
- Scott, A. J., & Storper, M. (2007). Regions, Globalization, Development. *Regional Studies*, 41(6 & 7), 579–593.
- Sharma, G., & Patil, G. R. (2024). Urban spatial structure and equity for urban services through the lens of accessibility. *Transport Policy*, 146, 72–90.
- Soja, E. (2000). *Postmetropolis: Critical studies of cities and regions*. Blackwell Publishing.
- Veneri, P. (2013). The identification of sub-centres in two Italian metropolitan areas: A functional approach. *Cities*, 31, 177–185.
- Xiao, W., Liu, W., & Li, C. (2022). Can the urban spatial structure accelerate urban employment growth? Evidence from China. *Growth and Change*, 53(4), 1668–1693.
- Yu, L., Zheng, W., Yu, T., & Wu, Y. (2021). How to identify urban centers/subcenters in metropolises? An analytical review. *Habitat International*, 115(102397), 1–10.